

Multiple Choice (10 marks)

1. Let `l = [1,2,3,4]`. What is `sum(l)` in Python 3?
 - (a) 4
 - (b) 1
 - (c) 10
 - (d) 8
2. In Python 3, let `r = lambda q: q * 2`. What is `r(3)`?
 - (a) 2
 - (b) 6
 - (c) 3
 - (d) 1
3. In Python 3, what is `['a', 'Chemistry', 0, 1][1]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
4. Which types of functions grow the slowest?
 - (a) $O(N^{1/2})$
 - (b) $O(N^{1/4})$
 - (c) $O(N^{1/N})$
 - (d) $O(N)$
5. Which of the following is correct about Python?
 - (a) It supports automatic garbage collection.
 - (b) It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.
 - (c) Both of the above.
 - (d) None of the above.

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, converting the list `l = [1,2,2,3,4]` to a set results in `{1,2,3,4}`, removing duplicates.
7. True or False: In Python 3, the function `long(x [,base])` is used to convert a string to an integer.
8. True or False: In Python 3, the function `min(l)` returns the largest element in the list `l`, which is 4 for the list `[1, 2, 3, 4]`.
9. True or False: In Python 3, the expression `['a', 'Chemistry', 0, 1][-3]` evaluates to `'Chemistry'`.
10. True or False: In Python 3, accessing `tuple[0]` results in an error when the tuple contains mixed data types.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the characteristics of phishing attacks and analyze why a legitimate email from a bank asking you to verify a transaction by calling an official number is least likely to indicate a phishing attempt.
12. Discuss how heuristic algorithms can enhance programs designed to find the shortest driving route between two locations, and analyze their advantages over traditional methods.

End of Paper